

SIT225 Data Capture Technologies

Assessment Pass Task 2

Research-Informed Use Case Design & Ethical Analysis

Estimated time: 12-15 hours

Last updated: 14 June 2026

Due: End of Week 5

Hurdle: This task contributes to the Learning Portfolio milestone for Week 5. Submission required to stay on track for unit completion.

I. Instructions and Conditions

This task is due in **Week 5 (Sunday)**, and you are expected to start early. If your first submission is incomplete or incorrect, you may resubmit after receiving feedback. There will be **no extensions**, and late submissions will not be accepted. If your submission is not correct and complete by the deadline, it will be marked as a **FAIL**. This task is a **hurdle requirement**, failure to submit a correct version on time will result in failing the unit.

All submissions will be checked for **plagiarism** and **GenAI** use. *Any GenAI usage should be acknowledged (see template below) and you must include, in appendix, the prompt, raw response, and your changelog how you have adopted the GenAI response in your work.* You must complete the work **independently**, and you must not share or show your work to anyone else, unless it is explicitly instructed in your task to do so. If you have questions or encounter problems, attend the on-campus or online classes, or use the **Discussion Board** on the unit site.

Per Deakin policy, include an Acknowledgements section:

"This assessment work was developed with assistance from the following approved Deakin genAI tools:

- Deakin GEM (accessed June 2026): Used to *<generate structured critique on protocol design and confounding variables (3 iterations)>*.
- Microsoft Copilot for web (accessed June 2026): Used to *<summarise ASHRAE comfort standards and identify relevant literature.>*

All genAI-generated suggestions were critically evaluated and integrated only where supported by evidence. The final design, decisions, and justifications are my own."

II.Task Overview

This task extends your first pass assessment by deepening your critical engagement with data capture design. You will:

1. Critically analyse the privacy and ethical implications of your initial use case
2. Review literature to identify research gaps and refine your problem
3. Design an enhanced protocol grounded in scholarly practice, including hypotheses and theoretical validation

This task assesses:

- ULO2: Apply professional and ethical standards to sustainable data collection [ethical analysis]
- ULO3: Design data capture protocols to meet user and functional requirements [research-informed design]
- ULO1: Process, analyse and interpret data using existing devices [predictive modelling & validation]

Note: Protocol execution and live data collection are **out of scope**. This is a design, literature review, and planning exercise.

III.Task Components

Part 1: Privacy & Data Lifecycle Analysis of First Assessment (500–600 words)

Critically reflect on the data capture task you completed in Assessment 1.

1.1 Data Minimisation Audit

State your original problem and sensors. Then answer:

- *What data did you capture?* (e.g., temperature every 10 seconds, GPS coordinates, accelerometer x/y/z)
- *How much was necessary?* Did you collect more than needed? (Justify or identify waste)
- *Sampling frequency:* Was every 10 seconds optimal, or could you have used 30 seconds without losing patterns?
- *Duration:* You collected for ~20 minutes. Would 10 minutes have revealed the same insight?

Example reflection:

“I captured accelerometer data (x, y, z) every 1 second for 30 minutes to detect standing vs. walking. Retrospectively, I could have sampled every 5 seconds without losing activity transitions, reducing storage by 80%. I also captured raw values (0–1023) when normalised values (0–1) would have sufficed.”

1.2 Data retention & Lifecycle

- *Where was data stored?* (Local CSV on laptop, Arduino cloud, Firebase?)
- *Who had access?* (Just you, shared with lab partners, uploaded to university server?)
- *Retention policy:* Did you delete data after analysis, or does it still exist?

- *Risks identified:* Could the data be re-identified? (e.g., accelerometer patterns + timestamp linked to your ID)

Action: Propose a data retention policy for your use case.

Example:

"Temperature data was stored locally on my encrypted laptop and deleted 48 hours post-analysis. No identifiable information was linked. If this were deployed at scale, data should be anonymised, retained for ≤ 30 days, and only aggregated statistics shared with facility managers."

1.3 Ethical Stakeholders & Consent

- *Who are affected by this data collection?* (e.g., you alone, roommates, building occupants)
- Did you obtain consent? Should you have?
- What are the potential harms if data were breached or misused?

Propose: A simple consent or privacy notice for your use case (template provided below).

Part 2: Literature Review & Research Gap Identification (1000–1200 words)

Conduct a structured literature review to contextualise your initial use case (assessment 1) and identify a research gap.

2.1 Literature search

Search for 2–3 peer-reviewed articles or credible practitioner reports (such as industry reports) related to your use case.

Search strategies:

- Google Scholar (scholar.google.com)
- Deakin Library: Use SIT225 reading list or search databases (e.g., IEEE Xplore, ACM Digital Library)
- Examples:
 - Room temperature & productivity: "thermal comfort occupancy sensing"
 - Soil moisture automation: "IoT soil moisture prediction"
 - Activity recognition: "smartphone accelerometer activity classification"

2.2 Synthesis & Gap Analysis

For each source, document:

Source	Problem Addressed	Methods Used	Data Capture	Key Finding	Limitation
Smith et al. (2022)	Does classroom temperature affect test performance?	Regression analysis	DHT22 + survey	2°C change = 5% score drop	Only 1 classroom; 1 semester
Jones & Lee (2023)	Real-time soil moisture prediction	LSTM neural network	Soil moisture + weather API	92% accuracy 7-day forecast	Requires 2 years historical data

2.3 Identify Your Research Gap

Write 300–400 words identifying *what has NOT been adequately addressed*:

- *Example*: "Smith et al. examined temperature–productivity in classrooms, but only in winter.
Gap: No study compares seasonal effects (summer cooling vs. winter heating) or includes occupancy data. Does the relationship change when the room is empty?"
- *Example*: "Current soil moisture systems use threshold-based triggers (e.g., water when <40%).
Gap: No study combines soil moisture + rainfall prediction + plant growth stage to optimise watering. This could reduce water use by 20%."

Part 3: Enhanced Use Case Design with Hypotheses (1200–1500 words)

Based on your literature review, design an improved use case that addresses the identified gap.

3.1 Refined Problem Statement

State your new problem clearly:

- *Original*: "Does room temperature affect focus?"
- *Refined*: "How do seasonal variation (summer vs. winter) AND occupancy patterns interact to predict optimal temperature, and can we reduce energy waste by 15% while maintaining comfort?"

3.2 Sensor & Data Capture Protocol Design

Specify all components. (Execution is out of scope; design only.)

Sensors:

- *Primary*: DHT22 (temperature, humidity)
- *Secondary*: PIR motion sensor (occupancy)
- *Tertiary*: Optional - light sensor (circadian impact) or CO₂ (air quality proxy)

Data Capture Protocol:

<i>Parameter</i>	<i>Value</i>	<i>Justification</i>
<i>Sampling frequency</i>	Every 5 min	Captures gradual thermal drift; 20% less data than every 1 min
<i>Collection duration</i>	3 months (winter, spring, summer)	Seasonal comparison; sufficient for trend detection
<i>Storage</i>	Firebase Realtime DB + local CSV backup	Cloud redundancy; allows real-time dashboard access
<i>Data fields</i>	timestamp, temp, humidity, occupancy, location (room ID), user_comfort_rating (optional)	Minimised: only captures occupancy state (binary), not identity
<i>Privacy: Anonymised</i>	Room "A" not linked to student name	Data deleted post-analysis; no personal identifiers

3.3 Hypotheses

State 2–3 testable hypotheses based on your literature review:

H1 (Main):

"Temperature–comfort relationship differs seasonally. In summer, occupants tolerate higher variance (22–26°C); in winter, tolerance is narrower (20–22°C)."

Operationalisation: Capture comfort ratings (1–5 scale) paired with temp/humidity readings. Test via linear regression: $\text{comfort} \sim \text{temperature} + \text{season} + \text{interaction}(\text{temperature} \times \text{season})$.

H2 (Secondary):

"Unoccupied rooms spend >60% of time at non-optimal temperatures due to lag in HVAC response. Smart scheduling based on PIR occupancy prediction could reduce energy by 12%."

Operationalisation: Compare actual temperature to setpoint; quantify time spent $>1^{\circ}\text{C}$ from target. Calculate hypothetical savings if HVAC activated 15 min before PIR detects occupancy.

H3 (Exploratory):

"Humidity levels correlate with reported discomfort (dry air in winter, high humidity in summer). Multi-sensor dashboards improve user awareness and reduce complaints by 20%."

Operationalisation: Collect humidity + weekly user feedback. Visualise on dashboard; measure satisfaction before/after dashboard rollout.

3.4 Theoretical Validation (Pre-Execution Checks)

Perform *desk-based validation* before live data collection. (No hardware needed.)

A. Literature-Based Validation

- Does your hypothesis align with existing theory?
 - *Example:* "Fanger's Predicted Mean Vote (PMV) model predicts comfort $\pm 2^{\circ}\text{C}$ around 22°C in winter. My H1 aligns with this."
 - Cite relevant frameworks or models from literature.

B. Statistical Feasibility

- Will your sample size support your analysis?
 - *Example:* "I'll collect 3 months $\times$ 1,440 readings/day = $\sim 130\text{k}$ data points. Linear regression requires $n \geq 30$; I have massive oversampling. However, temporal autocorrelation may require time-series methods (ARIMA, mixed effects)."
 - Identify statistical assumptions you'll need to test (normality, independence, linearity).

C. Sensor Validation

- Are your sensors fit for purpose?
 - *Example:* "DHT22 accuracy is $\pm 0.5^{\circ}\text{C}$. If comfort tolerance is $\pm 1^{\circ}\text{C}$, accuracy is sufficient. PIR sensor has 5–10 min warm-up; I'll discard first 10 readings."
 - Reference sensor datasheets (provided in course materials).

D. Data Quality Checks (Planned)

- How will you handle missing data?
 - *Example:* "If Firebase disconnects, local CSV backup triggers. Missing <5% of readings acceptable; >5% will flag that day for exclusion."
- Outlier strategy?
 - *Example:* "Temperature $> 40^{\circ}\text{C}$ or $< 5^{\circ}\text{C}$ flagged as sensor error (outside real-world range for indoor space). Will remove and interpolate if <1% of data."

3.5 Monitoring & Real-Time Notification Design (Optional)

If your use case benefits from live monitoring:

Dashboard Requirements:

- *Display:* Real-time temperature + humidity + occupancy status
- *Threshold alerts:* Alert if temp stays $> 25^{\circ}\text{C}$ for > 30 min (potential waste) or $< 18^{\circ}\text{C}$ (discomfort)
- *Platform:* Plotly Dash (Python) or Arduino Cloud dashboard
- *User notification:* Email/SMS/browser alert (consider privacy—inform users they're being monitored)

Example logic:

```
IF occupancy == OFF AND temperature != setpoint for 15 min:  
  ALERT: "Room A unoccupied, HVAC overworking. Estimated waste: $2/hr"  
ENDIF
```

```
IF occupancy == ON AND (temp < 20 OR temp > 26):  
  ALERT: "User feedback? Rate comfort (1–5)" [opt-in survey]  
ENDIF
```

3.6 Revised Data Lifecycle & Ethics (400–500 words)

Reflect on how your enhanced design addresses privacy concerns:

- *Data minimisation:* How does your revised protocol reduce unnecessary collection vs. the original?
- *Retention:* Propose a concrete retention policy (e.g., "aggregate to daily averages after 7 days; delete raw data after 3 months").
- *Consent:* Design a privacy notice for occupants (1–2 paragraphs).
- *Stakeholders:* Who benefits (facility managers, occupants, environment)? Any power imbalances?

Example privacy notice:

"This room uses motion sensors and temperature monitors to optimise comfort and energy use. Data is anonymised (no names, IDs). Raw readings are retained for 7 days, then aggregated to daily averages and retained for 90 days. Room occupants can request data deletion. Facility managers will receive weekly summaries only. Your comfort feedback (optional) is used to refine thresholds."

IV.Submission Checklist

Component	Format	Length	Notes
Week 3-5 activities	Word or PDF		Perform each week's activity and export PDF to finally combine together with other items mentioned below. Clearly set heading of each week and activity number.
Part 1: Privacy analysis	Reflective essay	800–1000 words	Data minimisation, retention, stakeholders, consent template
Part 2: Literature review	Report + synthesis table	1000-1200 words	2–3 sources, gap analysis, citations
Part 3: Enhanced Use Case	Design document	1200-1500 words	Problem, sensors, protocol table, 3 hypotheses, validation, ethics
Supporting materials	Appendices (optional)		Sensor datasheets, literature PDFs, privacy notice draft, statistical assumptions checklist
<i>Consolidate each part of the task into a single PDF (total word count ~3000-3700 words), append previous weeks' (week 3-5) activity sheets at the bottom of the PDF. There are no code or data to share for this assessment.</i>			

V.Grading Criteria (Pass/Fail)

Criterion	Pass	Notes
<i>Part 1 clarity</i>	Privacy audit is concrete (references original use case data); retention policy is specific; consent template is usable	Don't be vague (e.g., instead of saying "data is stored safely", you should say "data stored in local encrypted CSV, deleted after 48 hrs")
<i>Part 2 rigour</i>	2–3 credible sources found; gap clearly articulated; distinguishes gap from original problem	Scholarly articles preferred; grey literature (reports) acceptable if peer-reviewed equivalent unavailable
<i>Part 3 design</i>	Hypotheses are testable & grounded in literature; protocol is detailed & feasible; theoretical validation addresses ≥3 checks	Theory validation can be desk-based; no experiment needed

<i>Criterion</i>	<i>Pass</i>	<i>Notes</i>
	(stats, sensor, data quality)	
<i>Ethics integration</i>	Revised use case addresses identified privacy gaps; stakeholders articulated; consent/notification drafted	Show iterative improvement from Part 1 to Part 2+3
<i>Writing & structure</i>	Clear narrative; citations present; tables/diagrams used effectively	Professional but accessible; Australian English

Pass awarded when: All criteria met to satisfactory standard.

VI.Support Resources

- From course materials (Week 2–6): Sensor datasheets (DHT22, PIR, accelerometer), Firebase & Arduino Cloud setup guides, Plotly Dash templates, Data validation checklist.
- Predictive modelling: Scikit-learn logistic regression tutorial
- SIT225 reading list (recommended articles on IoT, privacy, data ethics)
- Literature sources: Google Scholar, IEEE Xplore, ResearchGate
- Privacy frameworks: GDPR articles 5–6 (data minimisation); ISO 27001 (data retention)

VII.Common Pitfalls to Avoid

- Skipping literature search; stating gap without evidence
- Hypotheses that are vague ("temperature affects comfort") or untestable
- Protocol design that ignores privacy (e.g., capturing user ID + biometric data)
- Theoretical validation that's superficial (e.g., "DHT22 is good" without referencing specs)
- No distinction between original use case (Part 1) and refined design (Part 3).